

GENERAL SPECIFICATION G7.17

Building Management System

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G7.17 BUILDING MANAGEMENT SYSTEM

G7.17.1 General

- (a) All works shall be provided by a single Building Management System (BMS) contractor.
- (b) The works shall be scheduled, coordinated, and interfaced with the associated work of the ACMV system and fire alarm system.
- (c) The BMS shall be as described by the Specifications and Drawings.

G7.17.2 Definitions

Definitions	Subject
Analog	A continuously variable system or value not having discrete levels. Typically exists within a defined range of limiting values
Binary	A two-state system where an “ON” condition is represented by one discrete signal level and an “OFF” condition is represented by a second discrete signal level
Control Sequence	A pre-programmed arrangement of software algorithms, logical computation, target values and limits as required to attain the defined operational control objectives
Programmable Logic Controllers (PLC)	Digital devices capable of being programmed to provide sequence control and continuous control inclusive of Proportional, Derivative and Integral control algorithms together with target values, limits, logical functions, arithmetic functions, constant values, timing considerations and the like
BMS Network	The total digital on-line real-time interconnected configuration of BMS digital processing units, workstations, panels, sub-panels, controllers, devices and associated elements individually known as network nodes. May exist as one or more fully interfaced and integrated sub-networks, LAN, WAN or the like.
Node	A digitally programmable entity on the BMS network
BMS Integration	The complete functional and operational interconnection and interfacing of all BMS work elements and nodes in compliance with all applicable codes, standards and ordinances so as to provide a single coherent BMS as required by the Specifications.
PC	Personal Computer from a recognized major manufacturer

- (a) The following abbreviations and acronyms may be used in describing the work of this Specifications:

Abbreviations/Acronyms	Subject
ADC	Analog to Digital Converter
AI	Analog Input
ANSI	American National Standards Institute
AO	Analog Output
ASCII	American Standard Code for Information Interchange
ASHRAE	American Society of Heating, Refrigeration and Air Conditioning Engineers
CPU	Central Processing Unit
DAC	Digital to Analog Converter
PLC	Programmable Logic Control
DI	Digital Input
DO	Digital Output
EEPROM	Electrically Erasable Programmable Read Only Memory
EMI	Electromagnetic Interference
FAS	Fire Alarm Detection and Annunciation System
GUI	Graphical User Interface
HOA	Hand-Off-Auto
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
I/O	Input/Output
IT	Information Technology
LAN	Local Area Network
LCD	Liquid Crystal Display
LED	Light Emitting Diode
MCC	Motor Control Centre
NC	Normally Closed
NO	Normally Open
OPC	OLE for Process Control
OWS	Operator Workstation
OAT	Outdoor Air Temperature
PC	Personal Computer

Abbreviations/Acronyms	Subject
RAM	Random Access Memory
RF	Radio Frequency
RFI	Radio Frequency Interference
RH	Relative Humidity
ROM	Read Only Memory
RTD	Resistance Temperature Device
SCADA	Supervisory Control and Data Acquisition
SPDT	Single Pole Double Throw
SPST	Single Pole Single Throw
XVGA	Extended Video Graphics Adapter
TBA	To Be Advised
TCP/IP	Transmission Control Protocol/Internet Protocol
TTD	Thermistor Temperature Device
UPS	Uninterruptible Power Supply
VAC	Volts, Alternating Current
VAV	Variable Air Volume
VDC	Volts, Direct Current
WAN	Wide Area Network

G7.17.3 General

- (a) The Building Management System (BMS) shall use an open architecture and fully support a multi-vendor environment. To accomplish this effectively, the BMS shall support open communication protocol standards and integrate a wide variety of third-party devices and applications.
- (b) The Building Management System shall consist of the following:
 - (i) Programmable Logic Controller(s);
 - (ii) Input/output Module(s);
 - (iii) Servers and Workstations;
 - (iv) Network processing, data storage and communications equipment; and
 - (v) Other components required for a complete and working BMS.

- (c) The system shall be modular in nature and shall permit expansion of both capacity and functionality through the addition of sensors, actuators, controllers and operator devices, while re-using existing controls equipment. Compliance with latest edition of ISO 9001 for quality assurance in design, development, production, installation and servicing.
- (d) System architectural design shall eliminate dependence upon any single device for alarm reporting and control execution. Compliance with EN50128/IEC62279 for software design, development and implementation.
 - (i) The failure of any single component or network connection shall not interrupt the execution of control strategies at other operational devices; and
 - (ii) The System shall maintain all settings and overrides through a system reboot.
- (e) System architectural design shall eliminate dependence upon any single device for alarm reporting and control execution.

G7.17.4 BMS Architecture

- (a) BMS Network
 - (i) The BMS system shall support high speed communication using Ethernet TCP/IP. The network communication shall be BacNet IP or Modbus TCP/IP.
 - (ii) The following features shall be supported by the BMS System:
 - 1) Open to third party devices supporting Modbus TCP/IP, BacNet IP, Ethernet IP, and OPC
 - 2) Embedded security features in compliance with IEC62443 standards
 - 3) Encrypted password access
 - 4) Strict supervision of firmware and software integrity
 - 5) Memory with error correction code
 - (iii) The PLC system shall support communications with third-party BMS SCADA platforms. The BMS SCADA platform software must support dual ring backbone network topology as well as ring and star topology to PLC's.
 - (iv) The BMS SCADA system provided shall be capable of duty / standby configuration regardless whether dual, star or ring network is implemented.
 - (v) The BMS SCADA system shall have the capability to communicate between distributed servers which allow the BMS SCADA workstations to communicate to multiple servers at the same time.
 - (vi) Industrial type OPC servers shall be provided if required to facilitate communications between different elements of the BMS and also with the Fire System.
- (b) BMS PLC
 - (i) The BMS PLC's shall support communications with third-party BMS SCADA platforms using at least one of the following protocols:
 - 1) Modbus TCP/IP;
 - 2) BacNet IP;

- 3) Ethernet/IP;
 - 4) Profinet;
 - 5) OPC.
- (ii) The Contractor shall provide an OPC servers, or other gateways required.
- (iii) Three licensed copies of PLC programming software shall be provided for all different types of installed BMS PLC's. This software shall be installed as follows:
- 1) One copy on BMS SCADA server machine;
 - 2) One copy on BMS maintenance Notebook computer; and
 - 3) One copy on another machine as instructed by the S.O..
- (iv) PLC I/O
- 1) PLC i/o shall be used to monitor and control inputs and outputs for use in the BMS PLC system.
 - 2) PLC I/O shall connect to the BMS PLC controllers via an Ethernet network.
 - 3) The I/O shall be assembled in BMS Control Panels as specified below.
- The Contractor shall provide the i/o required to monitor and control the building systems in accordance with the Drawings and Specifications.
- (c) Integration with mechanical plant shall be by one of the following methods
- (i) Hardwired
- 1) Analog and digital signal values shall be passed from one system to another via hardwired connections.
 - 2) There shall be one separate physical point on each system for each point to be integrated between the systems.
- (ii) Device Specific Protocol
- 1) The BMS system shall include appropriate hardware equipment and software to allow bi-directional data communications between the BMS system and equipment manufacturers' control panels. The BMS shall receive, react to, and return information from multiple building systems, including but not limited to the chillers, boilers, variable frequency drives, power monitoring system, etc.
 - 2) All data required shall be mapped into the BMS PLC.
 - 3) Point inputs and outputs from the third-party controllers shall have real-time interoperability with BMS software features such as: Control Software, Energy Management, Custom Process Programming, Alarm Management, Historical Data and Trend Analysis, Totalization, and Local Area Network Communications.
- (iii) BacNet IP/ Modbus TCP/IP Protocol Integration
- 1) The common protocol used between systems shall be BacNet IP / Modbus TCP/IP.

- 2) A complete Protocol Implementation Conformance Statement (PICS) shall be provided for all BacNet IP / Modbus TCP/IP system devices.
- 3) The ability to command, share point object data, change of state (COS) data and schedules between the host and BacNet IP / Modbus TCP/IP systems shall be provided.

G7.17.5 BMS SCADA User Interface

- (a) The system shall support user preferences in the following screen presentations:
 - Alarm;
 - Trend;
 - Display; and
 - Applications.
- (b) All controller software operating parameters shall be displayed for the operator to view/modify from the user interface. These include: set points, alarm limits, time delays, PID tuning constants, run-times, point statistics, schedules, and so forth.
- (c) The Operator Interface shall incorporate comprehensive support for functions including, but not necessarily limited to, the following:
 - (i) User access for selective information retrieval and control command execution;
 - (ii) Monitoring and reporting;
 - (iii) Alarm, non-normal, and return to normal condition annunciation;
 - (iv) Selective operator overrides and other control actions;
 - (v) Information archiving, manipulation, formatting, display and reporting;
 - (vi) BMS internal performance supervision and diagnostics;
 - (vii) On-line access to user HELP menus;
 - (viii) On-line access to current BMS as-built records and documentation; and
 - (ix) Means for the controlled re-programming, re-configuration of BMS operation and for the manipulation of BMS database information in compliance with the prevailing codes, approvals and regulations for individual BMS applications.
- (d) The system shall support a list of application programs configured by the users that are called up by the following means:
 - The Tools Menu;
 - Hyperlinks within the graphics displays; and
 - Key sequences.
- (e) The operation of the control system shall be independent of the user interface, which shall be used for operator communications only. Systems that rely on an operator workstation to provide execution of the sequences of operations shall not be acceptable.

(f) Alarms

- (i) Alarms shall be routed directly from Network to workstation and servers. It shall be possible for specific alarms from specific points to be routed to specific workstation and servers. The alarm management portion of the user interface shall, at the minimum, provide the following functions:
 - 1) Log date and time of alarm occurrence;
 - 2) Generate a “Pop-Up” window, with audible alarm, informing a user that an alarm has been received;
 - 3) Allow a user, with the appropriate security level, to acknowledge, temporarily silence, or discard an alarm;
 - 4) Provide an audit trail on hard drive for alarms by recording user acknowledgment, deletion, or disabling of an alarm. The audit trail shall include the name of the user, the alarm, the action taken on the alarm, and a time/date stamp;
 - 5) Provide the ability to direct alarms to an e-mail address or alphanumeric pager. This must be provided in addition to the pop up window described above. Systems that use e-mail and pagers as the exclusive means of annunciating alarms are not acceptable;
 - 6) Configuration of which PLC or IO modules offline alarms are seen by each user; and
 - 7) Any attribute of any object in the system may be designated to report an alarm.
- (ii) The BMS shall annunciate diagnostic alarms indicating system failures, PLC and IO modules offline and non-normal operating conditions.
- (iii) The BMS shall allow a minimum of 6 categories of alarm sound customizable through user defined wav. file format.
- (iv) The BMS shall annunciate application alarms as required.

(g) Reports and Summaries

- (i) Reports and Summaries shall be generated and directed to the user interface displays, with subsequent assignment to printers, or disk. As a minimum, the system shall provide the following reports:
 - All points in the BMS;
 - All points in each BMS application;
 - All points in a specific controller;
 - All points in a user-defined group of points;
 - All points currently in alarm;
 - All points locked out; and
 - All user defined and adjustable variables, schedules, interlocks and the like.

- (ii) Summaries and Reports shall be accessible via standard UI functions and not dependent upon custom programming or user defined HTML pages.
- (iii) Selection of a single menu item, tool bar item, or tool bar button shall print any displayed report or summary on the system printer for use as a building management and diagnostics tool.
- (iv) Provide the capability to view, command and modify large quantities of similar data in tailored summaries created online without the use of a secondary application like a spreadsheet. Summary definition shall allow up to seven user defined columns describing attributes to be displayed including custom column labels. Up to 100 rows per summary shall be supported. Summary viewing shall be available over the network using a standard Web browser.
- (v) Provide a focused set of reports that includes essential information required for effective management of energy resources within the facility.
- (vi) Reports shall be selectable by date, time, area and device. Each report shall include a colour visual summary of essential energy information.
- (h) Schedules
 - (i) A graphical display for time-of-day scheduling and override scheduling of building operations shall be provided. At a minimum, the following functions shall be provided:
 - 1) Weekly schedules;
 - 2) Exception Schedules; and
 - 3) Monthly calendars.
 - (ii) Weekly schedules shall be provided for each group of equipment with a specific time use schedule.
 - (iii) It shall be possible to define one or more exception schedules for each schedule including references to calendars
 - (iv) Monthly calendars shall be provided that allow for simplified scheduling of holidays and special days for a minimum of five years in advance. Holidays and special days shall be user-selected with the pointing device or keyboard and shall automatically reschedule equipment operation as previously defined on the exception schedules.
 - (v) Changes to schedules made from the User Interface shall directly modify the Network Automation Engine schedule database.
 - (vi) Schedules and Calendars shall comply with ASHRAE SP135/2008 BACnet Standard.
 - (vii) The Calendar object supports an option to add a reference to another Calendar Object that is designated to be the master for the facility. Any Supervisory and BAC calendars can be configured to reference a single master Global Calendar. Changes to the master global calendar are automatically synced with all calendars that are referenced.

- (viii) Selection of a single menu item or tool bar button shall print any displayed schedule on the system printer for use as a building management and diagnostics tool.
- (ix) Software shall be provided to configure and implement optimal start and stop programming based on existing indoor and outdoor environmental conditions as well as equipment operating history
- (i) Security/Passwords
 - (i) Multiple-level password access protection shall be provided to allow the user/manager to user interface control, display, and database manipulation capabilities deemed appropriate for each user, based on an assigned password.
 - (ii) Each user shall have the following: a user name (accept 24 characters minimum), a password (accept 12 characters minimum), and access levels.
 - (iii) The system shall allow each user to change his or her password at will and password about to expire.
 - (iv) When entering or editing passwords, the system shall not echo the actual characters for display on the monitor.
 - (v) A minimum of six levels of access shall be supported individually or in any combination as follows:
 - 1) Level 1 = View Data;
 - 2) Level 2 = Command;
 - 3) Level 3 = Operator Overrides;
 - 4) Level 4 = Database Modification;
 - 5) Level 5 = Database Configuration; and
 - 6) Level 6 = All privileges, including Password Add/Modify.
 - (vi) A minimum of 100 unique passwords shall be supported.
 - (vii) Operators shall be able to perform only those commands available for their respective passwords. Display of menu selections shall be limited to only those items defined for the access level of the password used to log-on.
 - (viii) Operators shall be further limited to only access, command, and modify those buildings, systems, and subsystems for which they have responsibility. Provide a minimum of 100 categories of systems to which individual operators may be assigned.
 - (ix) The system shall automatically generate a report of log-on/log-off and system activity for each user. Any action that results in a change in the operation or configuration of the control system shall be recorded, including: modification of point values, schedules or history collection parameters, and all changes to the alarm management system, including the acknowledgment and deletion of alarms.
 - (x) After successful login to the Site Management Portal (SMP) the last time and date that user name was previously logged in is shown on the screen.

- (xi) Each login attempt shall be recorded in the system Audit Log with the option to record the IP address of the PC that made the login.
- (j) Screen Manager
 - (i) The system will allow a customized image on the login screen (i.e. organization name, logo).
 - (ii) User View navigations can be displayed as either a set of tabs or a drop down list.
 - (iii) Allows user preference for assigning of a background colour for when an object is Out of Service which will enable the operator to quickly distinguish points that have been commanded to this state.
 - (iv) The User Interface shall be provided with screen management capabilities that allow the user to activate, close, and simultaneously manipulate a minimum of 4 active display windows plus a network or user defined navigation tree.
- (k) Dynamic Colour Graphics
 - (i) The graphics application program shall be supplied as an integral part of the User Interface. Browser or Workstation applications that rely only upon HTML pages shall not be acceptable.
 - (ii) The graphics applications shall include a create/edit function and a runtime function. The system architecture shall support an unlimited number of graphics documents (graphic definition files) to be generated and executed. The graphics shall be able to display and provide animation based on real-time data that is acquired, derived, or entered.
 - (iii) Graphics runtime functions – A maximum of 16 graphic applications shall be able to execute at any one time on a user interface or workstation with 4 visible to the user. Each graphic application shall be capable of the following functions:
 - 1) All graphics shall be fully scalable;
 - 2) The graphics shall support a maintained aspect ratio;
 - 3) Multiple fonts shall be supported;
 - 4) Unique background shall be assignable on a per graphic basis;
 - 5) The colour of all animations and values on displays shall indicate the status of the object attribute; and
 - 6) Graphics that represent buildings or systems shall allow natural links and transitions between related detailed tabular views of data that complement the graphic.
 - (iv) Operation from graphics – It shall be possible to change values (set points) and states in system-controlled equipment directly from the graphic.

- (v) Floor Plan graphics – The user interface shall provide graphic applications that summarize conditions on a floor. Floor plan graphics shall indicate thermal comfort using dynamic colours to represent zone temperature deviations from zone set point(s). Floor plan graphics shall display overall metrics for each zone in the floor.
- (vi) Aliasing – Many graphic displays representing part of a building and various building components are exact duplicates, with the exception that the various variables are bound to different field values. Consequently, it shall be possible to bind the value of a graphic display to aliases, as opposed to the physical field tags.
- (vii) Graphic editing tool – A graphic editing tool shall be provided that allows for the creation and editing of graphic files. The graphic editor shall be capable of performing/defining all animations and defining all runtime binding.
 - 1) The graphic editing tool shall provide a library of standard HVAC equipment, floor plan, lighting, security and network symbols.
 - 2) The graphic editing tool shall provide for the creation and positioning of library symbols by dragging from tool bars or drop-downs and positioning where required.
 - 3) The graphics editing tool shall permit the importing of AutoCAD drawings for use in the system.
 - 4) The graphic editing tool shall be able to add additional content to any graphic by importing images in the SVG, PNG or JPG file formats.
- (l) Historical Trending and Data Collection
 - (i) Each programmable logic controllers shall store trend and point history data for all analog and digital inputs and outputs, as follows:
 - 1) Any point, physical or calculated, may be designated for trending. Two methods of collection shall be allowed:
 - 2) Defined time interval; and
 - 3) Upon a change of value.
 - 4) Each Automation Engine shall have the capability to store multiple samples for each physical point and software variable based upon available memory, including an individual sample time/date stamp. Points may be assigned to multiple history trends with different collection parameters.
 - (ii) Trend and change of value data shall be stored within the engine and uploaded to a dedicated trend database or exported in a selectable data format via a provided data export utility. Uploads to a dedicated database shall occur based upon one of the following: user-defined interval, manual command, or when the trend buffers are full. Exports shall be as requested by the user or on a time scheduled basis.
- (m) Trend Data Viewing and Analysis
 - (i) Provide a trend viewing utility that shall have access to all database points.

- (ii) It shall be possible to retrieve any historical database point for use in displays and reports by specifying the point name and associated trend name.
 - (iii) The trend viewing utility shall have the capability to define trend study displays to include multiple trends
 - (iv) Displays shall be able to be single or stacked graphs with on-line selectable display characteristics, such as ranging, colour, and plot style.
 - (v) Display magnitude and units shall both be selectable by the operator at any time without reconfiguring the processing or collection of data. This is a zoom capability.
 - (vi) Display magnitude shall automatically be scaled to show full graphic resolution of the data being displayed.
 - (vii) The Display shall support the user's ability to change colours, sample sizes, and types of markers.
- (n) Database Management
- (i) Where a separate SQL database is utilized for information storage the System shall provide a Database Manager that separates the database monitoring and managing functions by supporting two separate windows.
 - (ii) Database secure access shall be accomplished using standard SQL authentication including the ability to access data for use outside of the Building Automation application.
 - (iii) The database managing function shall include summarized information on trend, alarm, event, and audit for the following database management actions:
 - Backup;
 - Purge; and
 - Restore.
 - (iv) The Database Manager shall support four tabs:
 - Statistics – shall display Database Server information and Trend, Alarm (Event), and Audit information on the Databases.
 - Maintenance – shall provide an easy method of purging records from the Server trend, alarm (event), and audit databases by supporting separate screens for creating a backup prior to purging, selecting the database, and allowing for the retention of a selected number of day's data.
 - Backup – Shall provide the means to create a database backup file and select a storage location.
 - Restore – shall provide a restricted means of restoring a database by requiring the user to log into an Expert Mode in order to view the Restore screen.
 - (v) The Status Bar shall appear at the bottom of all Database Manager Tabs and shall provide information on the current database activity. The following icons shall be provided:

- Ready;
 - Purging Record from a database;
 - Action Failed;
 - Refreshing Statistics;
 - Restoring database;
 - Shrinking a database;
 - Backing up a database;
 - Resetting internet information Services;
 - Starting the Device Manager;
 - Shutting down the Device Manager; and
 - Action successful.
- (vi) The Database Manager monitoring functions shall be accessed through the Monitoring Settings window and shall continuously read database information once the user has logged in.
- (vii) The System shall provide user notification via taskbar icons and e-mail messages when a database value has exceeded a warning or alarm limit.
- (viii) The Monitoring Settings window shall have the following sections:
- General – Shall allow the user to set and review scan intervals and start times;
 - Email – Shall allow the user to create and review e-mail and phone text messages to be delivered when a Warning or Alarm is generated;
 - Warning – shall allow the user to define the Warning limit parameters, set the Reminder Frequency, and link the e-mail message;
 - Alarm – shall allow the user to define the Alarm limit parameters, set the Reminder Frequency, and link the e-mail message; and
 - Database login – Shall protect the system from unauthorized database manipulation by creating a Read Access and a Write Access for each of the Trend, Alarm (Event) and Audit databases as well as an Expert Mode required to restore a database.
- (ix) The Monitoring Settings Taskbar shall provide the following informational icons:
- Normal – Indicates by colour and size that all databases are within their limits;
 - Warning - Indicates by colour and size that one or more databases have exceeded their Warning limit; and
 - Alarm - Indicates by colour and size that one or more databases have exceeded their Alarm limit.

- (x) The System shall provide user notification via Taskbar icons and e-mail messages when a database value has exceeded a warning or alarm limit.

G7.17.6 BMS SCADA Servers

- (a) Servers shall be industrial-grade, rack mounted and shall be configured to suit the installed software but as a minimum shall have the latest dual Eight-Core Intel Xeon processors with at least 3 GHz clock and expandable up to 2 Intel Xeon processors, 2x4MB cache memory, 16384MB DDR3 DIMM (FBD) 800MHz ECC memory, 730GB 15,000 RPM storage as RAID 6 SAS Solid State Drive, 25" Wide Screen Flat Panel LCD monitor (1920x1080 pixels non-interlacing), hot-pluggable redundant power supplies and cooling units, four Gigabit Ethernet network interfaces, high performance video graphics card with minimum of 128MB RAM, 16-bit sound card, speakers, CD-RW combination writer drive, DVD+RW dual layer writer with minimum write speed of 8x and with 100 blank 8.5GB DVD+R disks. Monitors, keyboards and ancillaries for servers shall be in the rack.
- (b) Each server shall as a minimum be supplied complete with suitable racks.

G7.17.7 BMS SCADA Workstations

- (a) The workstation design shall allow multiple, simultaneous (multi-tasking) window sessions, user terminal preference setup (e.g., colour choices, window sizes, etc.).
- (b) Workstations located in the control room shall be high-grade and shall be configured to suit the installed software but as a minimum shall have the latest Core 2 Duo Intel 64 bits processors with a minimum of 3.3GHz clock and 1333MHz front bus speed, 2MB L2 cache, 4GB DDR3 1033MHz memory, 500 GB 15,000 RPM SAS Solid State Drive, 24 speed DVD-ROM drive, dual 30" Wide Screen Flat Panel LCD monitors (1920x1080 pixels non-interlacing), high performance video graphics card with minimum of 512MB RAM, 16-bit sound card, speakers, and twin Gigabit Ethernet network interface and ancillaries.

G7.17.8 Field Devices

- (a) Input/Output Module
 - (i) The Input/Output Module (IOM) provides additional inputs and outputs for use in the PLC system.
 - (ii) The IOM shall support BacNet IP / Modbus TCP/IP protocol on the controller network.
 - (iii) The IOM shall be assembled in a plenum-rated plastic housing with flammability rated to UL94-5VB.
 - (iv) The IOM shall have a minimum of 4 points to a maximum of 17 points.
 - (v) The IOM shall support the following types of inputs and outputs:
 - 1) Universal Inputs - shall be configured to monitor any of the following:
 - Analog Input, Voltage Mode;
 - Analog Input, Current Mode;
 - Analog Input, Resistive Mode;

- Binary Input, Dry Contact Maintained Mode; and
- Binary Input, Pulse Counter Mode.
- 2) Binary Inputs - shall be configured to monitor either of the following:
 - Dry Contact Maintained Mode; and
 - Pulse Counter Mode.
- 3) Analog Outputs - shall be configured to output either of the following
 - Analog Output, Voltage Mode; and
 - Analog Output, current Mode.
- 4) Binary Outputs - shall output the following:
 - 24 VAC Triac.
- 5) Configurable Outputs - shall be capable of the following:
 - Analog Output, Voltage Mode; and
 - Binary Output Mode.
- (vi) The IOM shall include troubleshooting LED indicators to identify the following conditions:
 - 1) Power On;
 - 2) Power Off;
 - 3) Download or Startup in progress, not ready for normal operation;
 - 4) No Faults;
 - 5) Device Fault;
 - 6) Normal Data Transmission;
 - 7) No Data Transmission; and
 - 8) No Communication.
- (b) Networked Thermostat
 - (i) The networked thermostat shall be capable of controlling two-pipe fan coils, cabinet unit heaters or other similar equipment.
 - (ii) The Networked Thermostat shall support remote read/write and parameter adjustment from the web based User Interfaceable through BMSNetwork.
 - (iii) The Networked Thermostat shall include an intuitive User Interface providing plain text messages.
 - 1) Two line, 8 character backlit display;
 - 2) LED indicators for Fan, Heat, and Cool status;
 - 3) Five (5) User Interface Keys
 - Mode;
 - Fan;
 - Override;

- Degrees C/F; and
- Up/Down.
- 4) The display shall continuously scroll through the following parameters:
 - Room Temperature;
 - System Mode;
 - Schedule Status – Occupied/Unoccupied/Override; and
 - Applicable Alarms.
- (iv) The Networked Thermostat shall provide the flexibility to support any one of the following inputs:
 - 1) Integral Indoor Air Temperature Sensor;
 - 2) Duct Mount Air Temperature Sensor;
 - 3) Remote Indoor Air Temperature Sensor with Occupancy Override and LED Indicator; and
 - 4) Two configurable binary inputs.
- (v) The Networked Thermostat shall provide the flexibility to support any one of the following outputs:
 - 1) Three Speed Fan Control;
 - 2) Two On/Off;
 - 3) Two Floating; and
 - 4) Two Proportional (0 to 10V).
- (vi) The Networked Thermostat shall provide a minimum of six (6) levels of keypad lockout.
- (vii) The Networked Thermostat shall provide the flexibility to adjust the following parameters:
 - 1) Adjustable Temporary Occupancy from 0 to 24 hours;
 - 2) Adjustable cooling dead band from -17 °C to -15°C; and
 - 3) Adjustable cooling cycles per hour from 4 to 8.
- (viii) Where required by application and indicated on plans or room schedules provide the Networked Thermostat with an integral Passive Infra-Red (PIR) occupancy sensor.
- (ix) The Networked Thermostat shall employ non-volatile electrically erasable programmable read-only memory (EEPROM) for all adjustable parameters.

G7.17.9 Actuators and Operators

- (a) General Requirements
 - (i) Damper and valve actuators shall be electronic and/or pneumatic, as specified in the System Description section.
 - (ii) The manufacturer shall be ISO 9001 certified.

(b) Electronic Damper Actuators

(i) Spring Return Actuators

- 1) Regulatory Agency Listing: cULus, CSA C22.2 No. 24-93, and CE marked;
- 2) Direct-Coupled Design: Requires no crank arm or linkage for mounting to a shaft;
- 3) Coupling: toothed V-bolt clamp and nuts with toothed cradle;
- 4) Reversible Mounting: Provides either clockwise or counter clockwise operation;
- 5) Power Failure Operation: Mechanical spring return system drives load to the home position. Other forms of internal energy storage for power failure operation are not acceptable;
- 6) Motor Technology:
- 7) Modulating Types: Microprocessor-controlled Brushless DC motor; and
- 8) On/Off Types: DC brush motor.
- 9) Overload Protection: Electronic stall detection protects from overload at all angles of rotation without the use of end switches;
- 10) Enclosure Ratings:
- 11) NEMA type 2 / IP54 mounted in any orientation.
- 12) Double-Insulated construction: Eliminate the need for electrical ground wires;
- 13) Wiring: Integral cables with coloured and numbered conductors;
- 14) Sized for torque required to seal damper at load conditions;
- 15) Parallel Operation: Actuators shall be available that are capable of being mechanically or electrically paralleled;
- 16) Proportional actuators shall be user configurable without the use of external computer software or programming tools. Calibration, input signal range selection, and control logic reversal shall be selectable with an external mode selection switch;
- 17) Operating Temperature Range:
 - 7.9Nm. Torque and Below: 40°C to 60°C; and
 - 8.0Nm. Torque and above: 40°C to 55°C.
- 18) Power Requirements:
 - Modulating Types:
 - 3.0 Nm. Torque and Below: 5VA maximum
 - 7.9 Nm. to 2.1 Nm. Torque: 8VA maximum
 - 10 Nm. to 8.0 Nm. Torque: 10VA maximum
 - 10 Nm. to 20 Nm. Torque: 16VA maximum

- 2-Position Types:
 - 3.0 Nm. Torque and Below: 5VA maximum
 - 7.9 Nm. to 2.1 Nm. Torque: 7VA maximum
 - 8.0 Nm. to 20 Nm. Torque: 25VA maximum
- (ii) Non-Spring Return Actuators:
- 1) Regulatory Agency: UL Listed, CSA Certified, and CE marked;
 - 2) Direct-Coupled Design: Requires no crank arm or linkage for mounting to a shaft;
 - 3) Coupling:
 - Above 9.0 Nm.: toothed V-bolt clamp and nuts with toothed cradled; and
 - 9.0 Nm. and below: single cup-point set screw and toothed cradle.
 - 4) Overload Protection: Electronic stall detection or magnetic slip clutch protects from overload at all angles of rotation without the use of end switches;
 - 5) Minimum Enclosure Ratings:
 - Types with covered wiring terminals: NEMA type 2 / IP42 mounted in any orientation;
 - Types without covered wiring terminals: NEMA type 1 / IP30 or IP40; and
 - Types with integrated cables: NEMA 2 / IP42 mounted in any orientation.
 - 6) Sized for torque required to seal damper at load conditions;
 - 7) Parallel Operation: Actuators shall be available that are capable of being mechanically or electrically paralleled;
 - 8) Proportional actuators shall be user configurable without the use of external computer software or programming tools;
 - 9) Operating Temperature Range: -20°C to 50°C except for VAV and similar indoor applications in which case 0°C to 50°C is acceptable;
 - 10) Power Requirements: 24 V with models available for both 24 VAC and 24 VDC operation, maximum:
 - Above 9.0 Nm.: 7.5 VA at 24 VAC; and
 - 9.0 Nm. and below: 3.5 VA at 24VAC
 - 11) The manufacturer shall provide 5-year limited warranty from the date of sale covering defects in material or workmanship.

G7.17.10 Sensors and Transmitters

(a) General Requirements

- (i) Installation, testing, and calibration of all sensors, transmitters, and other input devices shall be provided to meet the system requirements.

(b) Temperature Sensors

(i) General

- 1) Sensors and transmitters shall be provided, as outlined in the input/output summary and sequence of operations.
- 2) The temperature sensor shall be of the resistance type and shall be either two-wire 1000-ohm nickel RTD, or two-wire 1000-ohm platinum RTD.
- 3) The following point types (and the accuracy of each) are required, and their associated accuracy values include errors associated with the sensor, lead wire, and A to D conversion:

Point Type	Accuracy
Chilled Water	$\pm .5^{\circ}\text{C}$.
Room Temp	$\pm .5^{\circ}\text{C}$.
Duct Temperature	$\pm .5^{\circ}\text{C}$.
All Others	$\pm .75^{\circ}\text{C}$.

(ii) Room Temperature Sensors

- 1) Room sensors shall be constructed for either surface or wall box mounting.
- 2) Room sensors shall have the following options when specified:
 - Setpoint warmer/cooler dial or reset slide switch providing a +3 degree (adjustable) range;
 - Individual heating/cooling setpoint slide switches;
 - A momentary override request push button for activation of after-hours operation; and
 - Analog thermometer.

(iii) Room Temperature Sensors with Integral Display

- 1) Room sensors shall be constructed for either surface or wall box mounting.
- 2) Room sensors shall have an integral LCD display and four button keypad with the following capabilities:
 - Display room air temperatures;

- Display and adjust room comfort setpoint;
- Display and adjust fan operation status;
- Timed override request push button with LED status for activation of after-hours operation;
- Display controller mode; and
- Password selectable adjustment of setpoint and override modes.

(iv) Outside Air Sensors

- 1) Outside air sensors shall be designed to withstand the environmental conditions to which they will be exposed. They shall also be provided with a solar shield.
- 2) Sensors exposed to wind velocity pressures shall be shielded by a perforated plate that surrounds the sensor element.
- 3) Temperature transmitters shall be of NEMA 3R (IP54) or NEMA 4 (IP65) construction and rated for ambient temperatures.
- 4) The outdoor sensor can be easily mounted on a roof, pole or side of a building utilizing its already assembled mounting bracket.
- 5) Outside Relative Humidity sensors 0-100% full range of accurate measurement. Operating temperature -20°C to 60°C.
- 6) Outside temperature sensors operating temperature range is -40 to 60°C, +/- .3°C.

(v) Duct Mount Sensors

- 1) Duct mount sensors shall mount in an electrical box through a hole in the duct and be positioned so as to be easily accessible for repair or replacement.
- 2) Duct sensors shall be insertion type and constructed as a complete assembly, including lock nut and mounting plate.
- 3) For outdoor air duct applications, a weatherproof mounting box with weatherproof cover and gasket shall be used.

(vi) Averaging Sensors

- 1) For ductwork greater in any dimension than 1.20m and/or where air temperature stratification exists, an averaging sensor with multiple sensing points shall be used.
- 2) For plenum applications, such as mixed air temperature measurements, a continuous averaging sensor or a string of sensors mounted across the plenum shall be used to account for stratification and/or air turbulence. The averaging string shall have a minimum of 4 sensing points per 3.60m long segment.
- 3) Capillary supports at the sides of the duct shall be provided to support the sensing string.

(vii) Humidity Sensors

- 1) The sensor shall be a solid-state type, relative humidity sensor of the Thin Film Capacitance or Bulk Polymer Design. The sensor element shall resist service contamination.
- 2) The humidity transmitter shall be equipped with non-interactive span and zero adjustments, a 2-wire isolated loop powered, 4-20 mA, 0-100% linear proportional output.
- 3) The humidity transmitter shall meet the following overall accuracy, including lead loss and Analog to Digital conversion. 3% between 20% and 80% RH @ 25°C unless specified elsewhere.
- 4) Outside air relative humidity sensors shall be installed with a rain proof, perforated cover. The transmitter shall be installed in a NEMA 3R (IP54) or NEMA 4 (IP65) enclosure with sealtite fittings.
- 5) A single point humidity calibrator shall be provided, if required, for field calibration. Transmitters shall be shipped factory pre-calibrated.
- 6) Duct type sensing probes shall be constructed of 304 stainless steel, and shall be equipped with a neoprene grommet, bushings, and a mounting bracket.

(viii) CO₂ Sensors

- 1) Where shown on the drawings, CO₂ sensors shall have the following features:
 - Jumper selectable: 0-20mA, 4-20mA & 0-10VDC output; and
 - Liquid Crystal Display.
- 2) The CO₂ sensors shall have the ability to monitor and output the following variables as required by the systems sequence of operations:
 - Zone carbon-dioxide.
- 3) The CO₂ shall transmit the information back to the controller via jumper selectable 0-20mA, 4-20mA & 0-10VDC output signals.
 - The CO₂ sensors shall provide a maximum output current of 25mA; Maximum output voltage of 12.5V; and
 - The CO₂ sensors shall be FCC compliant to CFR47 Part 15 subpart B Class A.
- 4) The CO₂ Sensors shall be available with
 - CO₂ response time (0-63%) of 1 minute;
 - Less than 0.083% of full scale/° temperature dependence of CO₂ output;
 - Long term CO₂ stability ±5% of full scale for 5 years;
 - CO₂ measurement accuracy of ± (40ppm + 2.0% of reading); and
 - CO₂ non-linearity of less than 1.0% of full scale.

- 5) The CO² Sensors may include the following items:
- Relay output module;
 - Liquid Crystal Display module; and
 - Analog temperature module with linear 0-10VDC output for 0-50°C.

G7.17.11 Differential Pressure Transmitters

- (a) General Air and Water Pressure Transmitter Requirements:
- (i) Pressure transmitters shall be constructed to withstand 100% pressure over-range without damage, and to hold calibrated accuracy when subject to a momentary 40% over-range input.
 - (ii) Pressure transmitters shall transmit a 0 to 5 VDC, 0 to 10 VDC, or 4 to 20 mA output signal.
 - (iii) Differential pressure transmitters used for flow measurement shall be sized to the flow sensing device and shall be supplied with Tee fittings and shut-off valves in the high and low sensing pick-up lines to allow the balancing Contractor and Owner permanent, easy-to-use connection.
 - (iv) A minimum of a NEMA 1 housing shall be provided for the transmitter. Transmitters shall be located in accessible local control panels wherever possible.
- (b) Low Differential Water Pressure Applications (0mm H₂O – 510mm H₂O)
- (i) The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of flow meter differential pressure or water pressure sensing points.
 - (ii) The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - 1) 0.25 - 510 mm H₂O input differential pressure range;
 - 2) 4-20 mA output;
 - 3) Maintain accuracy up to 20 to 1 ratio turndown; and
 - 4) Reference Accuracy: +0.2% of full span.
- (c) Medium to High Differential Water Pressure Applications (Over 540 mm H₂O)
- (i) The differential pressure transmitter shall meet the low pressure transmitter specifications with the following exceptions:
 - 1) Differential pressure range 250 mm H₂O to 300 PSI; and
 - 2) Reference Accuracy: +1% of full span (includes non-linearity, hysteresis, and repeatability).
 - (ii) Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.

- (d) Building Differential Air Pressure Applications (-25 to +25 mm H₂O)
 - (i) The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of differential pressure or air pressure sensing points.
 - (ii) The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - 1) -25 to +25 mm H₂O input differential pressure ranges. (Select range appropriate for system application);
 - 2) 4-20 mA output;
 - 3) Maintain accuracy up to 20 to 1 ratio turndown; and
 - 4) Reference Accuracy: +0.2% of full span.
- (e) Low Differential Air Pressure Applications (0 to 65 mm H₂O)
 - (i) The differential pressure transmitter shall be of industrial quality and transmit a linear, 4 to 20 mA output in response to variation of differential pressure or air pressure sensing points.
 - (ii) The differential pressure transmitter shall have non-interactive zero and span adjustments that are adjustable from the outside cover and meet the following performance specifications:
 - 1) (0.00 – 25.00 to 130) mm H₂O input differential pressure ranges. (Select range appropriate for system application.);
 - 2) 4-20 mA, 0-5 VDC, 0-10 VDC, output;
 - 3) Maintain accuracy up to 20 to 1 ratio turndown; and
 - 4) Reference Accuracy: +0.25%, or 0.5% of full span.
- (f) Medium Differential Air Pressure Applications (130 to 540 mm H₂O)
 - (i) The pressure transmitter shall be similar to the Low Air Pressure Transmitter, except that the performance specifications are not as severe. Differential pressure transmitters shall be provided that meet the following performance requirements:
 - 1) Zero & span: (c/o F.S./Deg. C): .04% including linearity, hysteresis and repeatability;
 - 2) Accuracy: 1% F.S. (best straight line) Static Pressure Effect: 0.5% F.S. (to 100 PSIG; and
 - 3) Thermal Effects: <+.033 F.S./Deg. C. over 4.40°C. to 38°C. (calibrated at 21°C.).
 - (ii) Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.

G7.17.12 Refrigerant Leak Detectors

- (a) The refrigerant leak detector shall be a standalone device and shall provide a SPDT output to directly energize the refrigeration room exhaust ventilation fans. The detector shall include a sensor or sensors connected to a control panel. Two relay contacts at the control panel shall provide trouble and alarm indication to the Facility Management System. The alarm relay contact shall also directly energize the exhaust fans.
- (b) The refrigerant leak detector shall sense the type of refrigerant used in the specified chillers. Multiple sensors shall be required to detect different refrigerants and/or provide proper sensing coverage for the area of the refrigeration room.

G7.17.13 Status and Safety Switches

- (a) General Requirements
 - (i) Switches shall be provided to monitor equipment status, safety conditions, and generate alarms at the BMS when a failure or abnormal condition occurs. Safety switches shall be provided with two sets of contacts and shall be interlock wired to shut down respective equipment.
- (b) Current Sensing Switches
 - (i) The current sensing switch shall be self-powered with solid-state circuitry and a dry contact output. It shall consist of a current transformer, a solid state current sensing circuit, adjustable trip point, solid state switch, SPDT relay, and an LED indicating the on or off status. A conductor of the load shall be passed through the window of the device. It shall accept over-current up to twice its trip point range.
 - (ii) Current sensing switches shall be used for run status for fans, pumps, and other miscellaneous motor loads.
 - (iii) Current sensing switches shall be calibrated to show a positive run status only when the motor is operating under load. A motor running with a broken belt or coupling shall indicate a negative run status.
- (c) Air Filter Status Switches
 - (i) Differential pressure switches used to monitor air filter status shall be of the automatic reset type with SPDT contacts rated for 2 amps at 120VAC.
 - (ii) A complete installation kit shall be provided, including: static pressure tops, tubing, fittings, and air filters.
 - (iii) Provide appropriate scale range and differential adjustment for intended service.
- (d) Air Flow Switches
 - (i) Differential pressure flow switches shall be bellows actuated mercury switches or snap acting micro-switches with appropriate scale range and differential adjustment for intended service.

- (e) Air Pressure Safety Switches
 - (i) Air pressure safety switches shall be of the manual reset type with SPDT contacts rated for 2 amps at 120VAC.
 - (ii) Pressure range shall be adjustable with appropriate scale range and differential adjustment for intended service.
- (f) Low Temperature Limit Switches
 - (i) The low temperature limit switch shall be of the manual reset type with Double Pole/Single Throw snap acting contacts rated for 16 amps at 120VAC.
 - (ii) The sensing element shall be a minimum of 5m in length and shall react to the coldest 200mm section. Element shall be mounted horizontally across duct in accordance with manufacturers recommended installation procedures.
 - (iii) For large duct areas where the sensing element does not provide full coverage of the air stream, additional switches shall be provided as required to provide full protection of the air stream.

G7.17.14 Control Relays

- (a) Control Pilot Relays
 - (i) Control pilot relays shall be of a modular plug-in design with retaining springs or clips;
 - (ii) Mounting Bases shall be snap-mount;
 - (iii) DPDT, 3PDT, or 4PDT relays shall be provided, as appropriate for application;
 - (iv) Contacts shall be rated for 10 amps at 120VAC; and
 - (v) Relays shall have an integral indicator light and check button.
- (b) Lighting Control Relays
 - (i) Lighting control relays shall be latching with integral status contacts;
 - (ii) Contacts shall be rated for 20 amps at 277 VAC;
 - (iii) The coil shall be a split low-voltage coil that moves the line voltage contact armature to the ON or OFF latched position;
 - (iv) Lighting control relays shall be controlled by:
 - 1) Pulsed Tri-state Output – Preferred method;
 - 2) Pulsed Paired Binary Outputs; and
 - 3) A Binary Input to the Facility Management System shall monitor integral status contacts on the lighting control relay. Relay status contacts shall be of the “dry-contact” type.
 - (v) The relay shall be designed so that power outages do not result in a change-of-state, and so that multiple same state commands will simply maintain the commanded state. Example: Multiple OFF command pulses shall simply keep the contacts in the OFF position.

G7.17.15 Electronic Signal Isolation Transducers

- (a) A signal isolation transducer shall be provided whenever an analog output signal from the BMS is to be connected to an external control system as an input (such as a chiller control panel) or is to receive as an input signal from a remote system.
- (b) The signal isolation transducer shall provide ground plane isolation between systems.
- (c) Signals shall provide optical isolation between systems.

G7.17.16 Electronic/Pneumatic Transducers

- (a) Electronic to Pneumatic transducers shall provide:
 - (i) Output: 3-15 PSIG;
 - (ii) Input: 4-20 mA or 0-10 VDC;
 - (iii) Manual output adjustment;
 - (iv) Pressure gauge; and
 - (v) External replaceable supply air filter.

G7.17.17 Thermostats

- (a) Electric room thermostats of the heavy-duty type shall be provided for unit heaters, cabinet unit heaters, and ventilation fans, where required. All these items shall be provided with concealed adjustment. Finish of covers for all room-type instruments shall match and, unless otherwise indicated or specified, covers shall be manufacturer's standard finish.

G7.17.18 System Tools

- (a) System Configuration Tool (SCT)
 - (i) The Configuration Tool shall be a software package enabling a computer SCADA platform to be used as a stand-alone engineering configuration tool for BMS Network.
 - (ii) The configuration tool shall provide an archive database for the configuration and application data.
 - (iii) The configuration tool shall have the same look-and-feel at the User Interface (UI) regardless of whether the configuration is being done online or offline.
 - (iv) The configuration tool shall include the following features:
 - 1) Basic system navigation tree for connected networks;
 - 2) Integration of BacNet IP / Modbus IP enabled devices;
 - 3) Customized user navigation trees;
 - 4) Point naming operating parameter setting;
 - 5) Graphic diagram configuration;
 - 6) Alarm and event message routing;
 - 7) Graphical logic connector tool for custom programming; and

- 8) Downloading, uploading, and archiving databases.
- (v) The configuration tool shall have the capability to automatically discover field devices on connected buses and networks.

G7.17.19 Local Control Panels

- (a) All control panels shall be factory constructed, incorporating the BMS manufacturer's standard designs and layouts. All control panels shall be UL inspected and listed as an assembly and carry a UL 508 label listing compliance. Control panels shall be fully enclosed, with perforated sub-panel, hinged door, and slotted flush latch.
- (b) In general, the control panels shall consist of the PLC controller(s), display module as specified and indicated on the plans, and I/O devices—such as relays, transducers, and so forth—that are not required to be located external to the control panel due to function. Where specified the display module shall be flush mounted in the panel face unless otherwise noted.
- (c) All I/O connections on the PLC controller shall be provided via removable or fixed screw terminals.
- (d) Low and line voltage wiring shall be segregated. All provided terminal strips and wiring shall be UL listed and provide adequate clearance for field wiring.
- (e) All wiring shall be neatly installed in plastic trays or tie-wrapped.
- (f) A 230-volt convenience outlet, fused on/off power switch, and required transformers shall be provided in each enclosure.

G7.17.20 Power Supplies

- (a) DC power supplies shall be sized for the connected device load. Total rated load shall not exceed 75% of the rated capacity of the power supply.
- (b) Input: 230 VAC, 50Hz.
- (c) Output: 24 VDC.
- (d) Line Regulation: +0.05% for 10% line change.
- (e) Load Regulation: +0.05% for 50% load change.
- (f) Ripple and Noise: 1 mV rms, 5 mV peak to peak.
- (g) An appropriately sized fuse and fuse block shall be provided and located next to the power supply.
- (h) A power disconnect switch shall be provided next to the power supply.

G7.17.21 Performance/Execution

- (a) BMS Specific Requirements
 - (i) Graphic Displays
 - 1) Provide a color graphic system flow diagram display for each system with all points as indicated on the point list. All terminal unit graphic displays shall be from a standard design library.

- 2) User shall access the various system schematics via a graphical penetration scheme and/or menu selection.
- (ii) Custom Reports:
 - 1) Provide custom reports as required for this project.
- (iii) Actuation / Control Type
 - 1) Primary Equipment
 - Controls shall be provided by equipment manufacturer as specified herein; and
 - All damper and valve actuation shall be electric.
- (iv) Air Handling Equipment
 - 1) All air handlers shall be controlled with a HVAC-PLC; and
 - 2) All damper and valve actuation shall be electric.
- (v) Terminal Equipment:
 - 1) Terminal Units (VAV, UV, etc.) shall have electric damper and valve actuation; and
 - 2) All Terminal Units shall be controlled with HVAC-PLC.

G7.17.22 Installation Practices

- (a) BMS Wiring
 - (i) All conduit, wiring, accessories and wiring connections required for the installation of the Building Management System shall be provided by the Contractor unless specifically shown on the Drawings. All wiring shall comply with the requirements of all local codes, unless specified otherwise in this section.
 - (ii) All BMS wiring materials and installation methods shall comply with BMS manufacturer recommendations.
 - (iii) The sizing, type and provision of cable, conduit, cable trays, and raceways shall be the design responsibility of the Contractor.
 - (iv) Provide for complete grounding of all applicable signal and communications cables, panels and equipment so as to ensure system integrity of operation. Ground cabling and conduit at the panel terminations. Avoid grounding loops.
- (b) BMS Line Voltage Power Source
 - (i) 240-volt AC circuits used for the Building Management System shall be taken from UPS Distribution Boards.
 - (ii) Circuits used for the BMS shall be dedicated to the BMS and shall not be used for any other purposes.
- (c) BMS Raceway
 - (i) All wiring shall be installed in conduit or raceway except as noted elsewhere in this specification. Minimum control wiring conduit size 12.7mm.

- (ii) Where it is not possible to conceal raceways in finished locations, surface raceway may be used as approved by S.O..
 - (iii) All conduits and raceways shall be installed level, plumb, at right angles to the building lines and shall follow the contours of the surface to which they are attached.
 - (iv) Flexible Metal Conduit shall be used for vibration isolation and shall be limited to 1m in length when terminating to vibrating equipment. Flexible Metal Conduit may be used within partition walls. Flexible Metal Conduit shall be UL listed.
- (d) Penetrations
 - (i) Provide fire stopping for all penetrations used by dedicated BMS conduits and raceways.
 - (ii) All openings in fire proofed or fire stopped components shall be closed by using approved fire resistive sealant.
 - (iii) All wiring passing through penetrations, including walls shall be in conduit or enclosed raceway.
 - (iv) Penetrations of floor slabs shall be by core drilling. All penetrations shall be plumb, true, and square.
- (e) BMS Identification Standards
 - (i) Node Identification. All nodes shall be identified by a permanent label fastened to the enclosure. Labels shall be suitable for the node location.
 - (ii) Cable shall be colour coded for easy identification and troubleshooting.
- (f) BMS Panel Installation
 - (i) The BMS panels and cabinets shall be located not less than 600mm from the bottom edge of the panel to the finished floor. Each cabinet shall be anchored per the manufacturer's recommendations.
 - (ii) The Contractor shall obtain the S.O.'s approval for all panel locations.
- (g) Input Devices
 - (i) All Input devices shall be installed per the manufacturer recommendation
 - (ii) Locate components of the BMS in accessible local control panels wherever possible.
- (h) HVAC Input Devices – General
 - (i) All Input devices shall be installed per the manufacturer recommendation
 - (ii) Locate components of the BMS in accessible local control panels wherever possible.
 - (iii) Input Flow Measuring Devices shall be installed in strict compliance with ASME guidelines affecting non-standard approach conditions.
 - (iv) Outside Air Sensors

- 1) Sensors shall be mounted on the North wall to minimize solar radiant heat impact or located in a continuous intake flow adequate to monitor outside air conditions accurately; and
 - 2) Sensors shall be installed with a rain proof, perforated cover.
- (v) Water Differential Pressure Sensors
- 1) Differential pressure transmitters used for flow measurement shall be sized to the flow-sensing device;
 - 2) Differential pressure transmitters shall be supplied with tee fittings and shut-off valves in the high and low sensing pick-up lines; and
 - 3) The transmitters shall be installed in an accessible location wherever possible.
- (vi) Medium to High Differential Water Pressure Applications (Over 530 mm H₂O):
- 1) Air bleed units, bypass valves and compression fittings shall be provided.
- (vii) Building Differential Air Pressure Applications (-25 mm H₂O to +25 mm H₂O)
- 1) Transmitters exterior sensing tip shall be installed with a shielded static air probe to reduce pressure fluctuations caused by wind; and
 - 2) The interior tip shall be inconspicuous and located as shown on the drawings.
- (viii) Air Flow Measuring Stations
- 1) Where the stations are installed in insulated ducts, the airflow passage of the station shall be the same size as the inside airflow dimension of the duct; and
 - 2) Station flanges shall be two inch to three inch to facilitate matching connecting ductwork.
- (ix) Duct Temperature Sensors
- 1) Duct mount sensors shall mount in an electrical box through a hole in the duct and be positioned so as to be easily accessible for repair or replacement;
 - 2) The sensors shall be insertion type and constructed as a complete assembly including lock nut and mounting plate;
 - 3) For ductwork greater in any dimension than 1.20m or where air temperature stratification exists such as a mixed air plenum, utilize an averaging sensor; and
 - 4) The sensor shall be mounted to suitable supports using factory approved element holders.
- (x) Space Sensors
- 1) Shall be mounted per ADA requirements; and
 - 2) Provide lockable tamper-proof covers in public areas and/or where indicated on the plans.
- (xi) Low Temperature Limit Switches

- 1) Install on the discharge side of the first water or steam coil in the air stream;
 - 2) Mount element horizontally across duct in a serpentine pattern insuring each square meter of coil is protected by 1 meter of sensor; and
 - 3) For large duct areas where the sensing element does not provide full coverage of the air stream, provide additional switches as required to provide full protection of the air stream.
- (xii) Air Differential Pressure Status Switches
- 1) Install with static pressure tips, tubing, fittings, and air filter.
- (xiii) Water Differential Pressure Status Switches
- 1) Install with shut off valves for isolation.
- (xiv) HVAC Output Devices
- 1) All output devices shall be installed per the manufacturers recommendation.
 - 2) Actuators: All control actuators shall be sized capable of closing against the maximum system shut-off pressure. The actuator shall modulate in a smooth fashion through the entire stroke. When any pneumatic actuator is sequenced with another device, pilot positioners shall be installed to allow for proper sequencing.
 - 3) Control Dampers: Shall be opposed blade for modulating control of airflow. Parallel blade dampers shall be installed for two position applications.
 - 4) Control Valves: Shall be sized for proper flow control with equal percentage valve plugs. The maximum pressure drop for water applications shall be 5 PSI. The maximum pressure drop for steam applications shall be 7 PSI.
 - 5) Electronic Signal Isolation Transducers: Whenever an analog output signal from the Building Management System is to be connected to an external control system as an input (such as a chiller control panel), or is to receive as an input a signal from a remote system, provide a signal isolation transducer. Signal isolation transducer shall provide ground plane isolation between systems. Signals shall provide optical isolation between systems

G7.17.23 Training

- (a) The Contractor shall provide the following training services:
 - (i) Two days of on-site orientation by a system technician who is fully knowledgeable of the specific installation details of the project. This orientation shall, at a minimum, consist of a review of the project as-built drawings, the BMS software layout and naming conventions, and a walk through of the facility to identify panel and device locations.

G7.17.24 Testing and Commissioning

- (a) Fully tested and commissioned in all aspects of the Building Management System work.
- (b) Factory Acceptance Test of all materials e.g. cables and equipment. Functional tests of the BMS. Thermal test on cabinets fully installed with BMS equipment.
- (c) Site Testing e.g. installation test, communication test, local point to point test, function test and system performance verification test.
- (d) Integrated Testing and Interfaces with electrical and mechanical equipment.
- (e) Acceptance Check Sheet
 - (i) Prepare a check sheet that includes all points for all functions of the BMS;
 - (ii) Submit the check sheet to the S.O. for approval; and
 - (iii) The S.O. will use the check sheet as the basis for acceptance with the Contractor.

END OF SECTION